MODESTA AWINO OTIENO

Software Developer

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PROFESSIONAL PROFILE

Software Developer with hands-on experience building web and backend applications using **Go, JavaScript, React, and modern development tools**. Currently developing software through practical, project-based training at **Zone01 Kisumu**, with experience in frontend development, backend services, REST APIs, databases, Git, Linux, and Docker. Experienced in developing practical applications that focus on security, reliability, usability, and problem solving. Interested in software engineering, full-stack development, backend systems, and building solutions that address real-world problems.

TECHNICAL SKILLS

Programming Languages	Go, C, JavaScript, HTML, CSS
Frontend Development	React, Vite, React Router, Responsive Web Development
Backend Development	Go HTTP Server, REST APIs, HTTP, HTML Templates, Node.js
Database & Data	PostgreSQL, Data Operations, Database Integration
Tools & Technologies	Git, GitHub, Linux, Docker, npm, Flutter
Software Development	Authentication, Role-Based Access Control, Audit Logging, API Integration, System Analysis and Design, Problem Solving

EXPERIENCE

Software Development Apprentice — Zone01 Kisumu

2026 – Present

- Develop software through practical, project-based programming and collaborative development.
- Build applications using Go, C, JavaScript, React, and other modern development technologies.
- Develop and work with HTTP servers, REST APIs, web interfaces, and application logic.
- Use Git and GitHub for version control and collaborative software development.
- Work in Linux-based development environments and debug applications through systematic problem solving.
- Apply software engineering principles to real-world development projects and collaborate through branches, commits, code reviews, and shared repositories.

ICT Support and Data Operations — Huduma Kenya

March 2026 – May 2026

- Provided technical support and troubleshooting for hardware and software issues.
- Diagnosed and resolved day-to-day ICT problems affecting users and computer systems.
- Assisted with network configuration and maintenance.
- Configured and maintained computer hardware, networks, and peripheral devices.
- Supported day-to-day ICT and data operations.

SELECTED PROJECTS

Smart Electoral Integrity System (SEIS)

React • JavaScript • Go • PostgreSQL • Docker • Git • GitHub

- Developed a secure demonstration platform focused on transparent, verifiable, and trustworthy digital election workflows.
- Built the frontend using React and Vite with reusable components and responsive interfaces.
- Designed authentication, role-based access control, audit logging, and results-verification workflows.
- Integrated a Go backend with PostgreSQL and used Docker to support development and deployment.

AXIS — Smart Irrigation & Water Management

Go • Web Development • Agricultural Data • REST API

- Developed a smart irrigation application for smallholder water-management decisions.
- Implemented crop water-use calculations using evapotranspiration concepts, crop growth stages, and irrigation efficiency.
- Built and tested the application as a practical solution for smallholder farming and deployed it for demonstration.

Tetris Optimizer

Go • Algorithms • File Parsing

- Implemented file parsing, input validation, tetromino representations, and placement logic.
- Developed algorithmic solutions for arranging pieces efficiently and handled invalid input according to project requirements.

Groupie Tracker

Go • REST APIs • HTML Templates • HTTP

- Built a Go web server and integrated external REST API data.
- Used HTML templates for dynamic pages and implemented artist information and relationship-based data presentation.
- Worked with server-side application logic, API communication, and error handling.

Lem-in

Go • Graphs • Algorithms • Parsing

- Parsed structured input data and built graph representations of rooms and connections.
- Implemented pathfinding logic and worked with graph-based algorithms and optimization.

EDUCATION

Rotary Vocational Training Centre — ICT

Currently Studying

Developing practical knowledge in information and communication technology, programming, computer systems, and software development.

PROFESSIONAL INTERESTS

Full-Stack Web Development • Backend Development • Software Engineering • Secure Application Development • APIs and Distributed Systems • Database Systems • Mobile Application Development • Problem Solving and Algorithms

LANGUAGE

English